Worksheet — Pendulum Tension Simulation

Use with *Simulation - Pendulum tension.html* · tension, circular motion (Block 5) and energy (Block 10)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/09-pendulum-tension** in a browser (offline: *Simulation - Pendulum tension.html*). Choose the object, its size, the string length and θ_0. ✋ **Hold aside** holds the bob still; ▶️ **Release** lets it swing. The newton meter at the top shows the tension.

Part 1 before the lab (≈ 15 min) · **Part 2** with a newton-meter pendulum (≈ 10 min) · **Part 3** homework (≈ 30 min).

Use g = 9.81 N kg^{-1}. L is measured to the **centre** of the bob.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. A 200 g mass is held still at 30° by a horizontal pull. Is T less than, equal to or more than mg?			
2. Released from 30°: T at the bottom of the swing compared with mg?			
3. Triple the string length, same 30° release. The peak tension?			
4. Double the mass. Peak tension? Period?			
5. Ping-pong ball or steel ball (same size): which keeps swinging longer?			

6. Draw the free-body diagram of the 200 g mass held still at 30°, and calculate the tension and the hand's pull.

Free-body diagram and working

Part 2 — During the lab: newton-meter pendulum (200 g)

situation	sim T / N	our reading / N	theory
at rest			mg
held at 20°			mg / cos 20°
held at 40°			mg / cos 40°
bottom of a swing from 40°			$mg(3 - 2\cos 40°)$

7. Why is the peak tension of a swing so hard to read on a real newton meter? How could you measure it better?

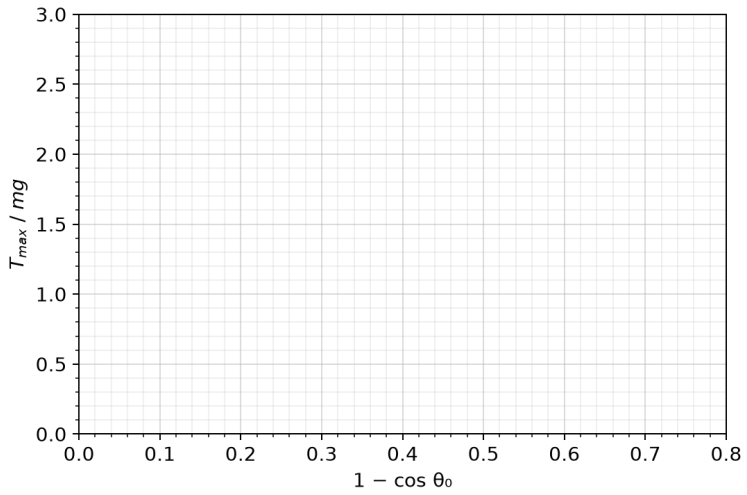
Part 3 — Homework

8. Use energy and the circular-motion equation to show that the tension at the bottom of a swing from θ_0 is $T = mg(3 - 2\cos\theta_0)$. Why does it not depend on L?

Derivation

θ_0 / °	10	20	30	40	50	60	70
T_{max} / N							
T_{max}/mg							
$1 - \cos\theta_0$							

9. Plot T_{max}/mg against $(1 - \cos\theta_0)$. Gradient = _____ Intercept = _____



10. Block 10 pendulum: L = 1.0 m released from 25°. Calculate the speed at the bottom and the tension there for a 1.0 kg bob. Then calculate the period for small swings.

Working

11. Explain why the balloon pendulum stops after a few swings while the steel ball keeps going.